

Influence of endogenous opiates on anterior pituitary function^{1,2}

DEAN A. VAN VUGT AND JOSEPH MEITES

Department of Physiology, Michigan State University, East Lansing, Michigan 48824

Although the discovery and isolation of endogenous opioid peptides (EOP) occurred only a few years ago, these compounds have generated great interest in their possible physiological functions in the body. Hughes et al. (23) isolated from the brain the endogenous opiates, methionine and leucine enkephalin (Met-Enk and Leu-Enk). A more potent opiate, beta-endorphin (β -End), was subsequently isolated from the pituitary, but also is present in the brain (3). Although much research has centered on the effects of these compounds on pain perception, behavior, and psychiatric diseases (16), their relatively high concentration in the hypothalamus and pituitary also suggest that they have endocrine functions. Several reports and a recent review (30) have appeared on their endocrine effects. In the present paper, we shall consider the evidence that they have a physiological role in regulating pituitary hormone secretion, and evaluate the reports on the mechanisms of their actions on the pituitary.

EFFECTS ON PRL AND GH

Morphine (MOR) was first reported in 1962 to induce release of prolactin (PRL) (29), based on its ability to initiate lactation in estrogen-primed rats. It was subsequently confirmed by radioimmunoassay (RIA) that MOR (26) and methadone (10) could elevate serum prolactin (PRL) in rats. In a recent study (6), we showed that MOR increased serum PRL concentrations and that this was prevented by concurrent administration of the opiate antagonist naloxone (NAL) (Table 1). In the same study, the three doses of MOR used produced even greater increases in serum growth hormone (GH) than in serum PRL. Other

workers also have reported that MOR can raise blood PRL and GH concentrations, and that this effect of MOR is blocked by concurrent administration of NAL (13, 26).

The EOP also can induce rapid increases in blood levels of PRL and GH. Table 1 shows that Met-Enk produced a modest increase in serum PRL. Furthermore, intraventricular injection of Met-Enk or β -End can raise serum PRL levels, but Met-Enk is much less effective than β -End. Dupont et al. (14) reported that in the rat 0.5 μ g of β -End produced a sevenfold increase in serum PRL, whereas 500 μ g of Met-Enk elicited only a fourfold rise in serum PRL. Cusan et al. (11) found that β -End was from 500 to 2,000 times more potent than Met-Enk in raising serum PRL and GH in the rat. Shaar et al. (46) reported that two enkephalin analogs, Met-Enk-amide and (D-Ala²)-Leu-Enk-amide, each stimulated PRL and GH release, and this could be blocked by injection of NAL. The enkephalin analog FK-33-824-

(DAMME) also was shown to increase PRL and GH in human subjects (48, 53).

An unexpected finding, first published by Shaar et al. (46) and our laboratory (6), was that NAL given alone significantly lowered serum PRL and GH concentrations in the rat. Table 1 shows that all three doses of NAL used significantly reduced serum PRL values below control values, whereas two of the three doses of NAL significantly decreased serum GH levels. Other investigators have

ABSTRACT

In general, the endogenous opioid peptides (EOP), morphine (MOR), and related drugs exert similar effects on acute release of pituitary hormones. Thus administration of opiates produces a rapid increase in release of prolactin (PRL), growth hormone (GH), adrenocorticotropin (ACTH), and antidiuretic hormone (ADH), and a decrease in release of gonadotropins and thyrotropin (TSH). Although not yet fully established, there is growing evidence that the EOP participate in the physiological regulation of pituitary hormone secretion. Thus naloxone (NAL), a specific opiate antagonist, has been shown to reduce basal serum levels of PRL and GH, and to elevate serum levels of LH and follicle stimulating hormone in male rats. Other reports have shown that NAL can inhibit the stress-induced rise in serum PRL, raise the castration-induced increase in serum LH to greater than normal castrate values, and counteract the inhibitory effects of estrogen and testosterone on LH secretion. Opiates appear to have no direct action on the pituitary, but there is evidence that they can alter activity of hypothalamic dopamine and serotonin in modulating secretion of pituitary hormones. —Van Vugt, D. A.; Meites, J. Influence of endogenous opiates on anterior pituitary function. *Federation Proc.* 39: 2533–2538; 1980.

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Abbreviations: β -End, beta-endorphin; EOP, endogenous opioid peptides; HAL, haloperidol; Leu-Enk, leucine enkephalin; Met-Enk, methionine enkephalin; MOR, morphine; NAL, naloxone; NE, norepinephrine; and PCPA, para-chlorophenylalanine.

TABLE 1. Effects of naloxone, morphine, and methionine enkephalin on serum hormones

Group	PRL	GH	LH	FSH	TSH
Controls					
0.87% NaCl	9.0 ± 0.4 ^a	131 ± 23	19 ± 1	341 ± 12	232 ± 33
Naloxone					
0.2 mg/kg	4.6 ± 0.4 ^b	77 ± 10 ^b	44 ± 5 ^b	361 ± 15	272 ± 40
Naloxone					
2.0 mg/kg	4.2 ± 0.3 ^b	103 ± 40	52 ± 8 ^b	490 ± 36 ^b	157 ± 22
Naloxone					
5.0 mg/kg	4.0 ± 0.3 ^b	48 ± 6 ^b	45 ± 8 ^b	446 ± 29 ^b	191 ± 25
Morphine					
2.0 mg/kg	10.1 ± 0.8	839 ± 172 ^b	8 ± 1 ^b	340 ± 12	122 ± 13
Morphine					
10.0 mg/kg	20.2 ± 2.3 ^b	1211 ± 185 ^b	9 ± 2 ^b	387 ± 23	77 ± 7 ^b
Morphine					
15.0 mg/kg	18.5 ± 1.2 ^b	1775 ± 172 ^b	13 ± 2	333 ± 13	71 ± 9 ^b
Met-Enk					
5.0 mg/kg	11.5 ± 0.6	215 ± 27 ^b	11 ± 2 ^b	314 ± 20	96 ± 17 ^b
NAL and MOR					
0.2 + 2.0 mg/kg	3.6 ± 0.4 ^b	383 ± 71 ^b	17 ± 3	384 ± 9	210 ± 13
NAL and MOR					
0.2 + 10.0 mg/kg	9.0 ± 1.1	902 ± 168 ^b	14 ± 4	369 ± 16	134 ± 17 ^b
NAL and Met-Enk					
0.2 + 5.0 mg/kg	5.4 ± 0.8 ^b	138 ± 37	17 ± 3	350 ± 18	191 ± 18

n = 10 per group. Values in ng/ml. ^a $\bar{x} \pm \text{SEM}$. Met-Enk, met-enkephalin; NAL, naloxone; MOR, morphine. ^b *P* < 0.05 as compared to controls. From (6), copyright 1977, by Bruni et al. Reprinted by permission.

confirmed that NAL can decrease serum PRL concentrations (33), but Martin et al. (28) reported that NAL did not alter pulsatile release of PRL and GH in the rat. The reasons for this discrepancy are not known at present, but we have confirmed that NAL can decrease serum PRL levels (54). In addition, we reported that NAL can inhibit stress-induced PRL release (54) (Fig. 1) and prevent growth of carcinogen-induced mam-

mary tumors in rats (2). Since these tumors are PRL-dependent, the effects of NAL on tumor growth probably results from inhibition of PRL release by NAL. Moreover, Dupont et al. (13) reported blockade of the suckling-induced rise in serum PRL by NAL in postpartum lactating rats, and also inhibition of the nocturnal rise of PRL in postmenopausal women.

There is evidence that stressful stimuli and high serum PRL and GH levels can alter EOP activity in the hypothalamus. Thus Akil et al. (1) found that intermittent foot shock increased opioid activity in the brain of mice as measured by bioassay. These results were confirmed by Wesche and Frederickson (57), using a specific RIA for Met-Enk. Rossier et al. (42) reported that the same stress increased serum β -End concentrations in rats. Cold exposure, which we showed increased serum PRL levels (35), was reported to elevate opiate activity in the pituitary (44). It is also of interest that stress was reported to elicit release of both ACTH and β -End into the blood of rats, and that these effects could be counteracted by administration of adrenal glucocorticoids (20). High blood PRL and GH concentrations may be able to decrease hypothalamic EOP levels, as indicated by a recent study showing that a pituitary

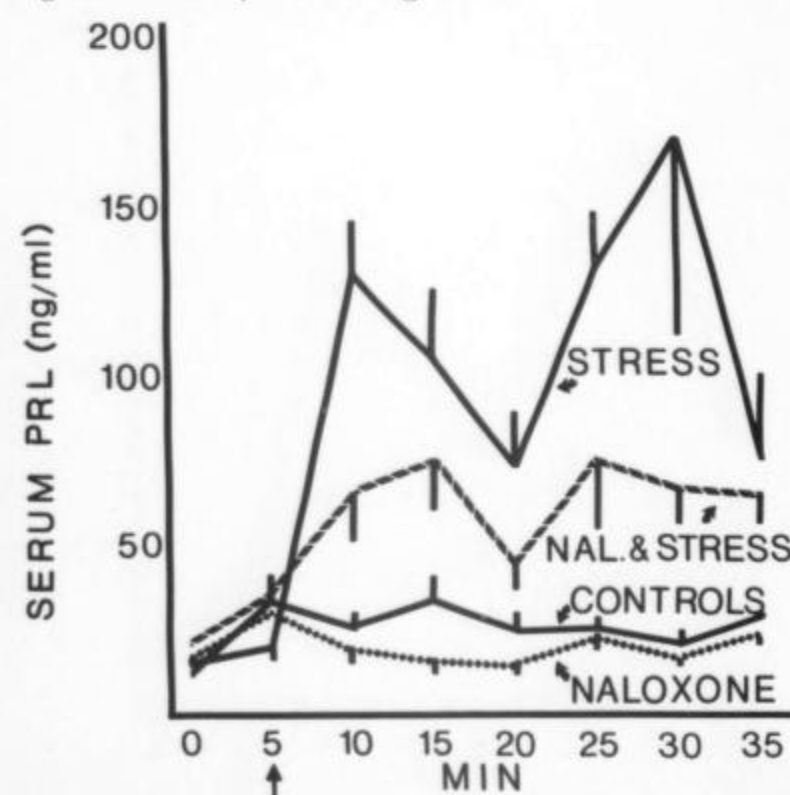
tumor graft that secreted large amounts of PRL and GH, reduced hypothalamic β -End content in female rats (37).

EFFECTS ON LH AND FSH

Barracough et al. (4) in 1955 reported that MOR injected during the critical period of proestrus prevented ovulation in rats. More recently, Pang et al. (38) demonstrated that this was due to inhibition of the proestrous luteinizing hormone (LH) surge, and that NAL counteracted the inhibitory effects of MOR on ovulation and LH release. Electrical stimulation of the median eminence also blocked the antioviulatory action of MOR, suggesting that MOR inhibits the release, but not necessarily the synthesis, of luteinizing hormone-releasing hormone (LHRH) (45). Cicero et al. (7) found that MOR and methadone reduced serum LH, (but not serum follicle stimulating hormone (FSH)) concentrations in male rats. We also observed that acute MOR treatment reduced serum LH, but had no effect on serum FSH concentrations in male rats, and that a single intraperitoneal injection of Met-Enk significantly reduced serum LH concentrations, while it had no effect on serum FSH levels (Table 1). Administration of NAL together with Met-Enk blocked the inhibitory effects of Met-Enk on LH secretion. Stubbs et al. (48) reported that in human subjects the Enk analog DAMME, given intravenously, significantly reduced serum LH concentrations; however, serum FSH values were only slightly depressed. NAL partially reversed the effects of DAMME on LH and FSH release.

The first evidence that the EOP may inhibit LH and FSH secretion was our finding that NAL alone significantly increased basal serum concentrations of both hormones in male rats (6). Table 1 shows that all three doses of NAL significantly elevated serum LH, and the two highest doses also raised serum FSH above control levels. When multiple injections of NAL were given to male rats soon after castration, serum LH levels were elevated above normal castrate levels, whereas MOR partially inhibited the postcastration rise of serum LH (55) (Fig. 2). Blank et al. (5) and Ieiri et al. (in press) reported

Figure 1. Effects of naloxone (NAL) on restraint stress induced prolactin (PRL) release. Blood was collected through an atrial cannula. The arrow indicates time of immobilization, and vertical bars represent the standard error of mean (SEM) (*n* = 4). The dose of NAL used was 0.2 mg/kg body weight. From (54), copyright, 1978, by Van Vugt et al.



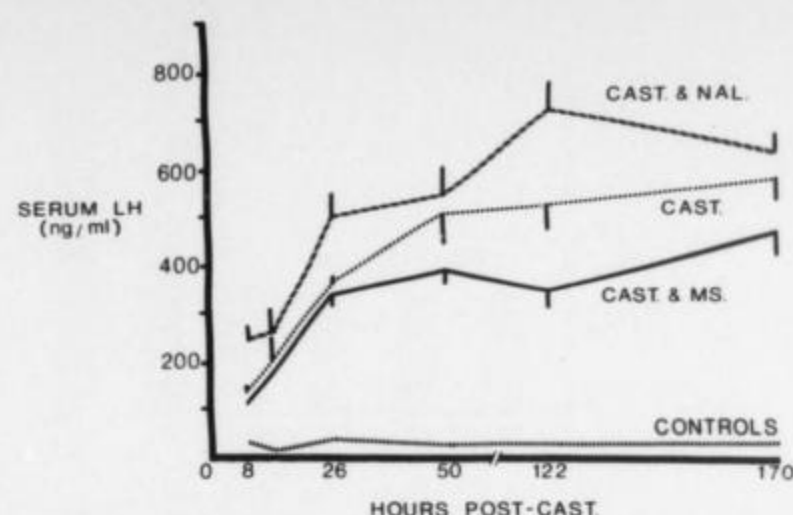


Figure 2. Effects of morphine sulfate (MS) and naloxone (NAL) on the post-castration rise of luteinizing hormone (LH). Castrated male rats were injected every 6 hr with either MS (10 mg/kg), NAL (0.2 mg/kg), or 0.87% saline. Blood was collected by orbital sinus puncture 6 hr after the previous injection. The vertical bars indicate SEM ($n = 10$). From (55), by Van Vugt et al.

that NAL inhibited LH release in prepubertal female rats, but had relatively little ability to do so in prepubertal male rats. In mature rats, NAL increased LH as well as FSH release (6).

TSH, ACTH AND POSTERIOR PITUITARY HORMONES

Acute administration of MOR has been reported to decrease thyrotropin (TSH) release and reduce thyroid function (6, 25). We also observed that MOR significantly decreased serum TSH levels, and that the inhibitory effect of MOR was partially blocked by NAL (Table 1). It can be seen that acute injection of Met-Enk also significantly reduced serum TSH values as compared to controls, and that this effect was counteracted by NAL. NAL alone had no significant effect on serum TSH values.

Acute MOR administration has the opposite effect on ACTH and adrenal corticosterone release. A single injection of MOR was reported to induce ACTH and adrenal cortical hormone release in the rat, mouse, and dog (18, 36). Injection of MOR into the arcuate nucleus produced a prolonged release of ACTH (17), whereas chronic MOR administration inhibited ACTH and adrenal cortical hormone release (36).

MOR apparently stimulates vasopressin (ADH) and inhibits oxytocin release. Neurosecretory material was reduced in the supraoptic and paraventricular nuclei and increased in the posterior pituitary by a single injection of MOR (39), suggesting that MOR stimulates release of oxytocin and ADH. However, Haldar

and Sawyer (21) recently reported that MOR inhibited oxytocin release. β -End, like MOR, was shown to increase ADH release in the rat (56). Conversely, opiate antagonists were reported to inhibit ADH release (34). Rossier et al. (41) found that placement of lesions in the paraventricular and supraoptic nuclei dramatically reduced Leu-Enk content in the intermediate lobe of the pituitary. They also found that salt loading reduced intermediate lobe concentration of the Leu-Enk to approximately 37% of control values. Leu-Enk cell bodies are present in the supraoptic and paraventricular nuclei, and fibers from these neurons parallel magnocellular neurosecretory axons. These studies indicate that Leu-Enk probably is involved in control of ADH release through an interaction with the magnocellular neurosecretory axons.

MECHANISMS OF OPIATE ACTION

Little is yet known about the mechanisms by which MOR and the EOP influence pituitary hormone secretion. It is generally agreed that MOR and the EOP do not act directly on the pituitary. Shaar et al. (46) observed no effect on the *in vitro* release of PRL or GH when enkephalin analogs or MOR were added to an incubation medium containing rat anterior pituitary tissue. Grandison and Guidotti (19) also reported that addition of naltrexone to an incubation medium containing rat pituitaries had no effect on PRL release. Lien et al. (24) reported slight stimulation of PRL release from monolayer cultures of rat pituitary cells by addition of Met-Enk or Leu-Enk, but this was not confirmed by Rivier et al. (40). In addition, it has been shown that the opiates do not interfere with the action of LHRH on pituitary LH release (8, 9), or of TRH on TSH release, and of TSH on thyroid function (47). Most of the current evidence indicates that the opiates act via the hypothalamus to influence pituitary function.

The hypothalamus contains hypophysiotropic hormones that are transported via the hypothalamo-pituitary portal vessels to regulate anterior pituitary hormone secretion. The chemical structures of three of

these hormones are known: thyrotropin-stimulating hormone (TRH) stimulates TSH and PRL and, under some conditions, GH secretion; LHRH stimulates LH and FSH secretion; and somatostatin inhibits GH, TRH-induced TSH secretion, insulin, glucagon, and gastrin secretion.

The hypothalamus also contains biogenic amines and other neurotransmitters that can influence the secretion of pituitary hormones, presumably by regulating release of the hypothalamic hypophysiotropic hormones into the portal vessels. Most is known about the roles of dopamine (DA), norepinephrine (NE), and serotonin (5-HT) on anterior pituitary function (32). In general, the evidence to date indicates that DA inhibits PRL, TSH, and ACTH release, it stimulates GH release, and it may either inhibit or stimulate LH and FSH release. Norepinephrine (NE) stimulates LH, FSH, TSH, and GH release, inhibits ACTH release, and may either increase or decrease PRL release. Serotonin increases PRL, GH, and ACTH release, and has been reported to either depress or promote gonadotropin and TSH release. Only DA has been shown to act directly on the anterior pituitary to influence (inhibit) release of a hormone (PRL), but it has no direct effect on release of any other anterior pituitary hormone. The other biogenic amines have no direct action on the anterior pituitary.

Ferland et al. (15) reported that Met-Enk decreased DA turnover in the median eminence, suggesting a mechanism by which opiates can stimulate PRL release. We similarly found that β -End decreased DA turnover in the median eminence, as shown in Fig. 3. We obtained further evidence that opiates decrease hypothalamic DA activity through the use of centrally active drugs. Thus the stimulatory effects of MOR on PRL secretion were inhibited by L-dopa, the precursor of DA, and by piribedil, a DA agonist (Table 2). Amineptine, a DA reuptake inhibitor, was prevented by MOR from reducing serum PRL levels, presumably by decreasing the amount of DA available at the amineptine site of action. When MOR was administered together with a dose of haloperidol (HAL) that effectively blocked DA receptors on the pituitary, MOR was unable to further

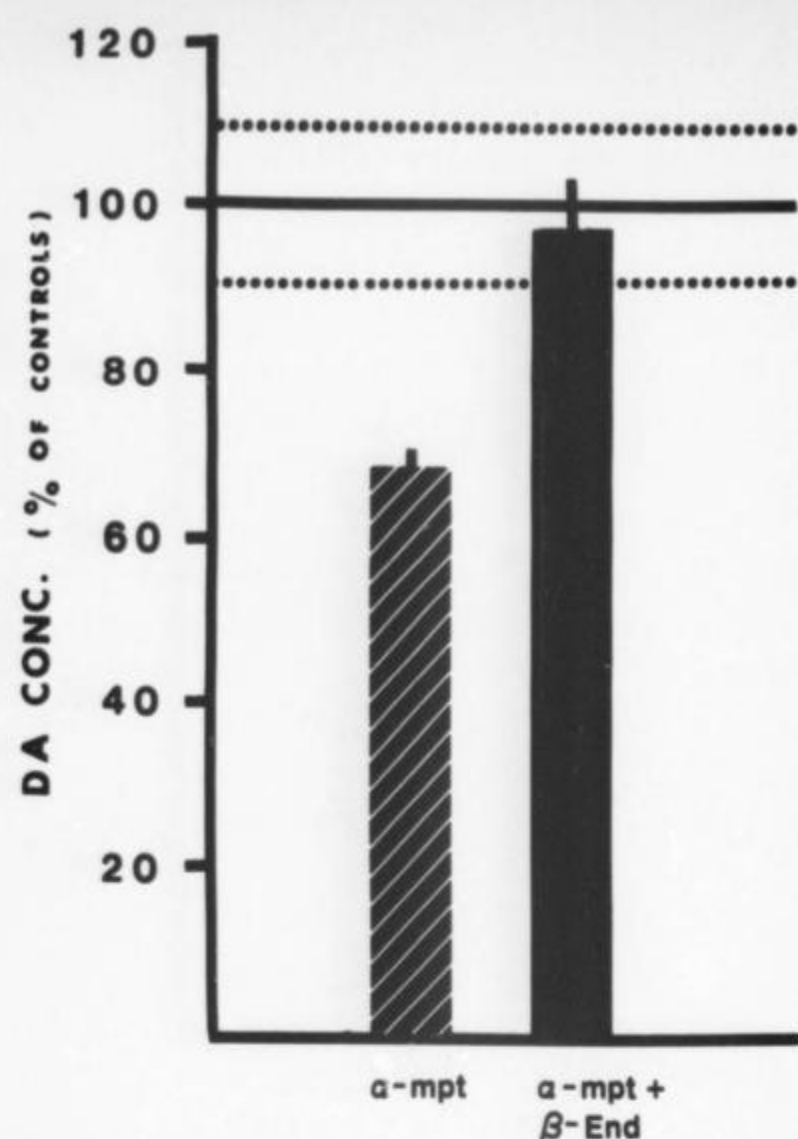


Figure 3. Effects of β -End on median eminence (ME) DA turnover. Since there was no significant difference in the ME dopamine concentration of rats treated with 0.87% NaCl and rats treated with β -End (131.4 ± 12.67 vs. 118.3 ± 18.1 ng/mg protein), these values were combined. The mean was set at 100% and is represented by the solid horizontal line. The dotted horizontal lines represent ± 1 SEM. The solid vertical bar represents ME DA concentration (122.2 ± 7.4 ng/mg protein) 1 hr after α -methyl-*p*-tyrosine (α -mpt) and β -End administration. The hatched bar represents DA concentration (85.8 ± 2.7 ng/mg protein) in the ME 1 hr after α -mpt injection. Both values are presented as % of controls. The vertical lines represent $\pm$ SEM based on 6 determinations. DA concentration in the ME of rats treated with α -mpt and rats treated with α -mpt together with β -End were significantly different ($P < 0.05$) from each other, as determined by Student's *t*-test for unpaired data.

increase serum PRL concentrations. However, when subeffective doses of HAL were injected together with a subeffective dose of MOR (5 mg/kg), a synergistic effect on PRL release was observed (Fig. 4). This synergistic action may indicate that the combination of the two drugs was able to reduce DA activity sufficiently to elevate serum PRL values. These results have recently been confirmed by Van Loon et al. (51), who reported that β -End reduced ventral hypothalamic DA turnover. They also found that the DA agonist apomorphine blocked the stimulatory action of β -End on PRL secretion.

Opiate depression of hypothalamic DA activity cannot explain its action

on release of GH. A decrease in hypothalamic DA should result in a reduction rather than in an increase in GH release, since DA stimulates GH release (32). Acute MOR administration was reported to increase serotonin (5-HT) turnover in the brain of rats (22, 58), and we observed that MOR increased 5-HT activity in the hypothalamus, as indicated by use of drugs (31). Van Loon et al. (50) observed that 5-HT turnover was increased in the hypothalamus by β -End administration. An increase in hypothalamic 5-HT turnover could explain opiate stimulation of GH release and inhibition of gonadotropin release, although the effects of 5-HT on GH release are still controversial (32). Dupont et al. (13) found that a reduction of 5-HT by parachlorophenylalanine (PCPA) partially inhibited the stimulatory effects of MOR on PRL secretion. However, Taché et al. (49) were unable to alter the PRL response to several different opiates with PCPA. MOR was reported to decrease NE concentrations in the whole brain and hypothalamus of rats (12, 52), which may account in part for the decrease in gonadotropin release by MOR.

Opiates do not appear to interfere with the actions of hypothalamic hypophysiotropic hormones on pituitary hormone release. Martin et al. (27) reported that MOR did not block the inhibitory effects of somatostatin

TABLE 2. Effects of dopaminergic drugs on morphine-induced increase in serum prolactin

	0 hr	1 hr	2 hr
Controls			
0.87% NaCl	11 $\pm$ 1 ^a	17 $\pm$ 3	13 $\pm$ 3
L-Dopa			
50 mg/kg	15 $\pm$ 4	3 $\pm$ 1	6 $\pm$ 2
Piribedil			
3 mg/kg	15 $\pm$ 3	3 $\pm$ 1	8 $\pm$ 1
Amineptine			
20 mg/kg	16 $\pm$ 4	6 $\pm$ 1	5 $\pm$ 1
Haloperidol			
1 mg/kg	14 $\pm$ 3	37 $\pm$ 3	33 $\pm$ 3
Morphine			
10 mg/kg	18 $\pm$ 3	36 $\pm$ 5	17 $\pm$ 7
L-Dopa + MOR	13 $\pm$ 3	6 $\pm$ 2	7 $\pm$ 2
PIR + MOR	16 $\pm$ 2	4 $\pm$ 1	4 $\pm$ 2
AMIN + MOR	16 $\pm$ 4	15 $\pm$ 2	5 $\pm$ 1
HAL + MOR	15 $\pm$ 2	36 $\pm$ 2	34 $\pm$ 2

^a $\bar{x} \pm$ SEM. MOR, morphine; PIR, piribedil; AMIN, amineptine; HAL, haloperidol. *n* = 6/group.

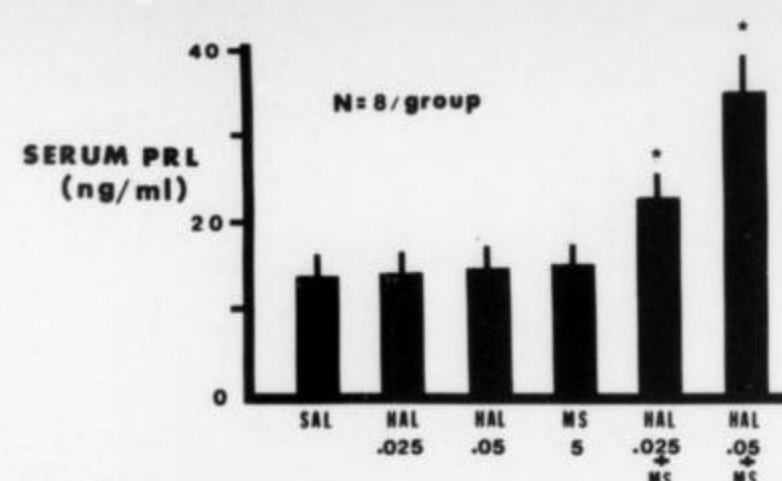


Figure 4. Effects of haloperidol (HAL) and morphine sulfate (MS) on serum prolactin (PRL) concentrations. Vertical bars represent SEM. The asterisks represent $P < 0.05$ as compared to controls. Number below each treatment indicates dose of drug (mg/kg).

on GH release. Chronic methadone therapy was reported not to alter the action of TRH on TSH release (47), and Cicero et al. (8) found that MOR administration had no effect on LHRH stimulation of LH release. There also is evidence that the opiates do not interfere with the action of TSH on thyroid function (25). Rotsztein et al. (43) recently reported that DA stimulated the release of LHRH from incubated hypothalamic tissue, and that this was inhibited by Met-Enk. We have mentioned the possibility that the EOP also may themselves directly influence release of hypothalamic hypophysiotropic hormones (30). It is obvious that further investigations are necessary to elucidate the mechanisms of opiate action on pituitary hormone secretion.

CONCLUSIONS

The discovery of the endogenous opioid peptides (EOP) has added a new dimension to studies on the control of pituitary functions. That the EOP can alter pituitary secretion is now firmly established, but further work is necessary to establish their physiological importance in regulating pituitary functions and to determine their mechanisms of action. Preliminary evidence indicates that the opiates can alter the activity of several hypothalamic neurotransmitters, particularly dopamine and serotonin. There also is the possibility that the EOP may act directly on the hypothalamic hypophysiotropic hormones to increase or decrease their release into the pituitary portal vessels. There are some preliminary indications that endocrine and other physiological stimuli may alter the hypotha-

lamic content of EOP, and thereby modulate pituitary hormone secretion. These and other problems of EOP-endocrine interactions remain to be investigated. **EP**

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